ECE501 - DIGITAL IMAGE PROCESSING

SAR-Optical Feature Fusion for Crop-Type Classification

REPORT - WEEK 7

Name	Enrollment No.
Srushti Gamit	AU2440205
Dhruvi Chavda	AU2440236
Chaithanya Jamula	AU2440097

GitHub Repository:
https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

1) Work done this week:

- Completed Sentinel-1 SAR time-series statistical analysis for all four pilot fields from Germany and Uruguay.
- Extracted VV and VH backscatter statistics, including mean, median, standard deviation, minimum and maximum values.
- Verified the SAR statistics and confirmed that all 214 acquisitions passed the quality checks.
- Generated SAR time-series graphs for VV and VH values for all four fields.

- Combined SAR backscatter data with Sentinel-2 optical NDVI data using a 3-day date-matching tolerance.
- Generated combined comparison graphs for NDVI, VV and VH across all four fields.
- Performed Pearson correlation analysis between NDVI and SAR VV/VH values.
- Generated eight scatter plots and a correlation summary CSV.

2) Work Planned for Next Week:

- Perform statistical significance testing using p-values for the calculated Pearson correlations.
- Review and finalize SAR–Optical fusion results for all four pilot fields.
- Complete any remaining analysis, validation and visualizations required for the project.
- Prepare final comparison tables and interpret the results.
- Organize the datasets, Python scripts, CSV files and generated graphs.
- Complete the final project report, presentation and documentation.
- Review the complete workflow and prepare the final project submission.

3) Results:

- Successfully processed 214 Sentinel-1 SAR acquisitions across four pilot fields.
- Generated 14 SAR time-series graphs.
- Successfully matched 126 out of 134 optical observations with SAR observations.
- Generated three combined comparison graphs for NDVI, VV and VH.
- Completed Pearson correlation analysis for all four fields.

Pearson Correlation Results:

Field	NDVI vs VV	NDVI vs VH

Germany Rapeseed 2019	0.7739	0.8104
Germany Wheat 2019	-0.4391	0.1669
Uruguay Soybean 2021	0.6026	0.7103
Uruguay Soybean 2020	0.3719	0.4368

- Generated eight correlation scatter plots with linear trend lines.
- Saved the correlation summary CSV and all generated graphs successfully.

4) Challenges Faced:

- Handling a large number of Sentinel-1 SAR acquisitions and downloading them within Google Earth Engine export limits.
- Maintaining consistent SAR image dimensions, CRS and spatial resolution.
- Matching SAR and optical observations because their acquisition dates were different.
- Handling cloud-affected Sentinel-2 images and applying quality filtering before NDVI calculation.
- Understanding the differences in correlation patterns between the four pilot fields.

5) References:

1. **Google Earth Engine – Sentinel-1 GRD Dataset:** **https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD**

2. **Google Earth Engine – Sentinel-2 SR Harmonized Dataset: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED**
3. **Google Earth Engine Documentation: https://developers.google.com/earth-engine**
4. **Rasterio Documentation: https://rasterio.readthedocs.io/**
5. **Pandas Documentation: https://pandas.pydata.org/docs/**
6. **SciPy Documentation: https://docs.scipy.org/doc/scipy/**